NutriGoal

API Integration Documentation

This document explains how NutriGoal integrates external APIs using HTTP requests to retrieve, send, parse, and display dynamic data.

1. Primary External API: Groq API

API Name: Groq API (OpenAI-compatible Chat Completions)

Purpose: AI-generated meal recommendations and meal plans based on user profile (activity, goal, health, allergies, budget).

Endpoint: POST https://api.groq.com/openai/v1/chat/completions

Method: HTTP POST

Implemented in: src/lib/groqMealPlan.js, src/lib/recommendCategories.js

Features added:

- AI meal plan generation (1-day / 7-day)
- Category matching with explanations
- Dynamic recipes (ingredients, steps, prices, calories)
- Local fallback if API fails

Sample JSON response (simplified):

```
{
  "choices": [{
    "message": {
      "content": "{\"nutritionTip\":\"Drink water.\",\"recommendations\":[{\"meal\":\"Lunch\"}]}"
    }
  }]
}
```

2. Backend / Database API: Supabase

API Name: Supabase API (Auth + Database + Storage)

Purpose: User authentication, meal plan CRUD storage, and profile avatar uploads.

Endpoints: Project URL from VITE_SUPABASE_URL; Auth; table saved_meal_plans; Storage bucket

Methods: HTTP GET (retrieve meal plans), write/POST-like ops (insert/update/auth), storage upload

Implemented in: supabaseClient.js, mealPlanDb.js, Login/Register, MealLayout

Features added: Login/register, save/read/update/delete meal plans, profile photo upload

Sample retrieved data:

```
[{
  "id": "uuid-row-id",
  "created_at": "2026-08-10T08:00:00.000Z",
  "is_saved": true,
  "plan": {"categoryTitle": "High Protein", "budget": 500, "people": 2}
}]
```

3. Supporting API: OTP (Google Apps Script)

API Name: Custom OTP Backend (Apps Script Web App)

Purpose: Send/verify OTP during registration/login.

Endpoint: VITE_OTP_SCRIPT_URL

Method: HTTP POST (sendOtp, verifyOtp)

Implemented in: src/lib/otpClient.js

Sample response: {"ok": true, "message": "OTP sent"}

How Requirements Are Met

Requirement	How NutriGoal Meets It
Integrate >= 1 API	Groq + Supabase + OTP
HTTP GET	Supabase fetchMealPlans() retrieves plans
HTTP POST	Groq generation; OTP; auth writes
Parse JSON	response.json() + meal plan parsers
Display in UI	Meal Plan, categories, History/Saved
Loading/errors	Spinners, try/catch, fallback
Hide API keys	.env via import.meta.env (not hard-coded)

Security Notes

API keys are not hard-coded. They are loaded from .env (VITE_GROQ_API_KEY, VITE_SUPABASE_*, VITE_OTP_SCRIPT_URL). .env is gitignored; submit .env.example only.

Screenshot Guide

1. API request implementation — fetch() in groqMealPlan.js or otpClient.js
2. Retrieved data displayed — generated meal plan / categories / History
3. JSON response sample — DevTools Network tab (Groq or Supabase)